TÉCNICO
UNIVERSIDADE
DE LISBOA

CONTAINER SHIPS · SHIP AND OCEAN SYSTEMS DESIGN

Container Ship Design Notes

A presentation to guide you through the design manual *Container Ships · Sizing from Bays, Rows and Tiers*.

Alvariño et al. (1997) §2.9 and §3.8.6 · Papanikolaou (2026) §1.3.7, §2.15–2.19 · M. Ventura, Container Carriers · Design manual, Steps 0 to 8 and the worked example

Tiago Santos, Luís Batista

Instituto Superior Técnico / University of Lisbon

PART A

Container Carriers

History, types and sizes, the growth of the largest ships, the rules that shape them and the less common configurations.

Historical Note: the Ideal X

- The first container carrier, the **Ideal X**, opened a line from Newark, New Jersey, to Houston, Texas, on 26 April 1956.
- A converted tanker, its deck reinforced to carry **58 containers of 35 ft**.
- The idea came from Malcom McLean (1914–2001), owner of trucking companies.
- McLean founded Sea-Land in 1960; Maersk bought it in 1999.
- The Ideal X was in service from 1956 to 1965.

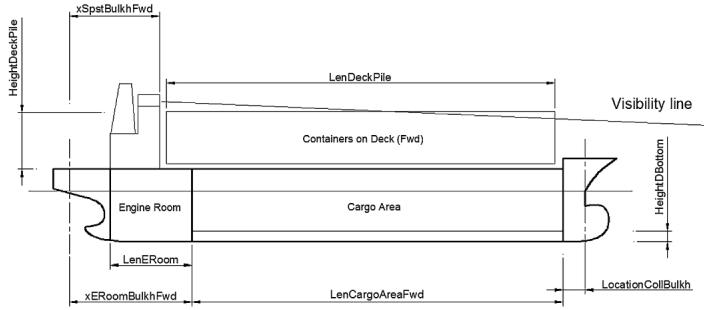


Types and Sizes

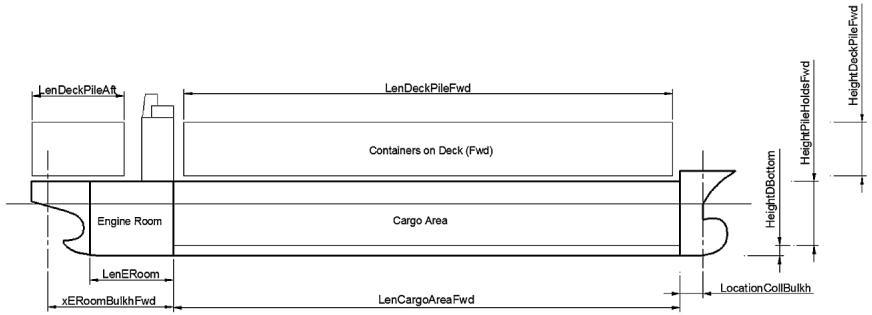
Type	What sets it	From Ventura’s notes
Feeder	own cranes on many ships	superstructure compact and aft; e.g. 900 TEU
Panamax	beam of the old Panama locks, about 32.2 m	up to 4,800 TEU; 13 rows on deck
Post-Panamax	wider than the old locks	e.g. 8,000–8,200 TEU, B 42.8 m
VLCC	very large container carriers	e.g. 7,200–9,600 TEU
ULCS	more than 10,000 TEU	Suez draught 16.4 m; Malacca 21 m, about 18,000 TEU

After the enlargement of the Panama Canal locks, the beam allows 22 containers across the deck instead of 13.

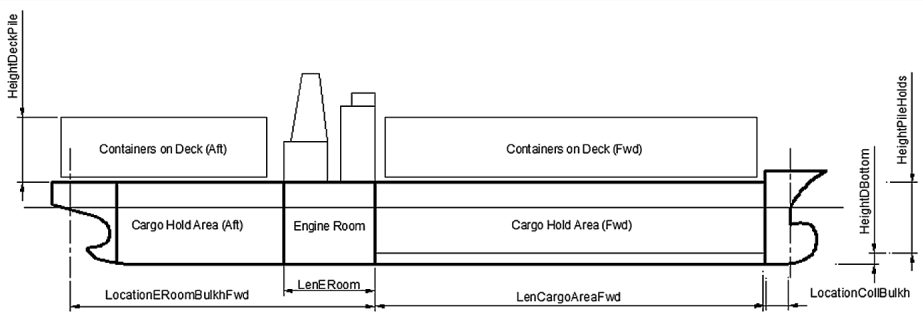
Four Typical Layouts



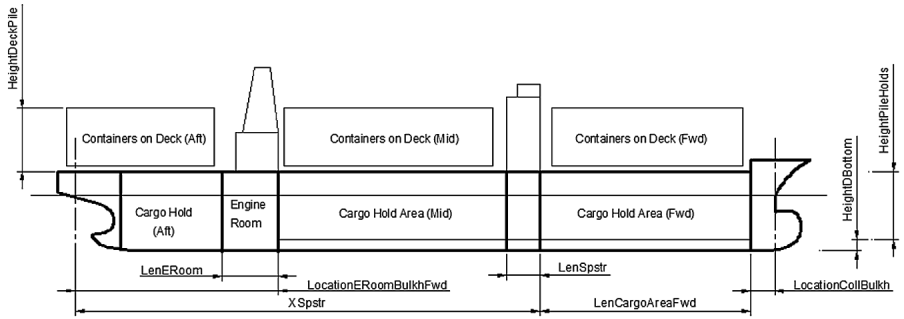
Feeder. Engine room and bridge aft; two cargo areas.



Panamax. Engine room and bridge aft; three cargo areas.



Post-Panamax. Engine room and bridge amidships; four areas.



Malaccamax. Engine room aft, bridge amidships; six areas.

The Four Layouts in Service



Feeder. OOCL · 900 TEU



Panamax. APL · up to 4,800 TEU



Post-Panamax. COSCO · 8,204 TEU · L 334 m



Malaccamax. MSC · 19,224 TEU · L 395.4 m

Growth of Size: 2006 to 2009



Emma Maersk, 2006

Item	Value
Capacity	11,000 TEU, 1,000 reefers
L × B × T	397.7 × 56.4 × 15.54 m
Power; speed	80,960 kW; 27 kn
Across the deck	22 rows; bridge amidships



MSC Danit, 2009

Item	Value
Capacity	14,028 TEU
L × B × T	365.5 × 51.2 × 16.0 m
Power; speed	72,240 kW; 25 kn
Across the deck	20 rows

Maersk Triple-E: the Two-Island Design

- 18,000 TEU, 16 % more than Emma Maersk; L 400 m, B 59 m, T 14.5 m
- U-shaped hull and **23 rows** across, one more than Emma: about 1,500 more containers
- Bridge and accommodation **5 bays forward**, engine room and funnel **6 bays aft**
- About 250 more containers before the bridge without losing visibility, about 750 more behind it







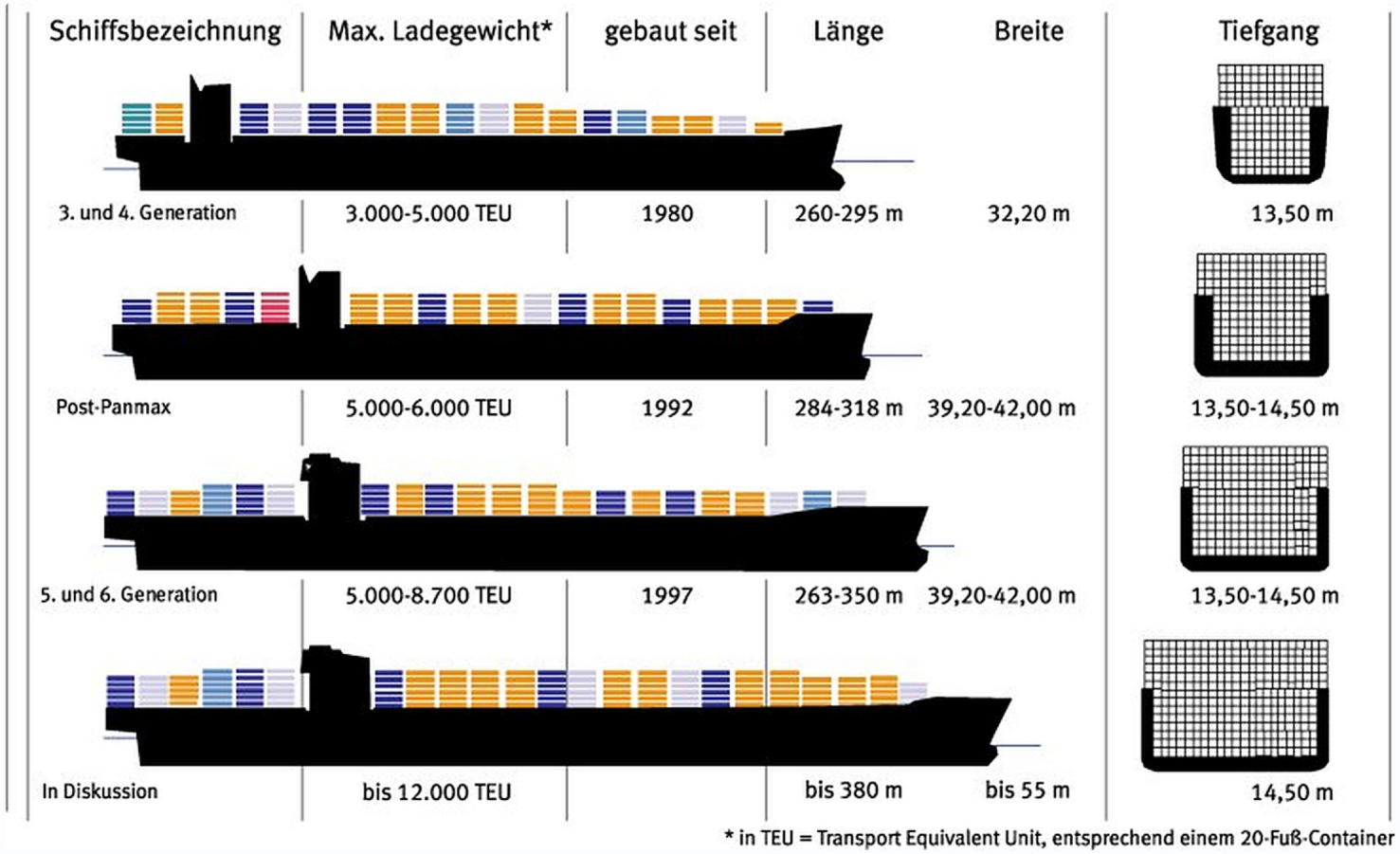
The Largest Ships Today

Ship	Year	TEU	L × B	Across deck
MSC Oscar	2015	19,224	395.4 × 58.6 m	23
MSC Gulsun	2019	23,756	399.9 × 61.4 m	24
HMM Algeciras	2020	23,964	399.9 × 61.0 m	—

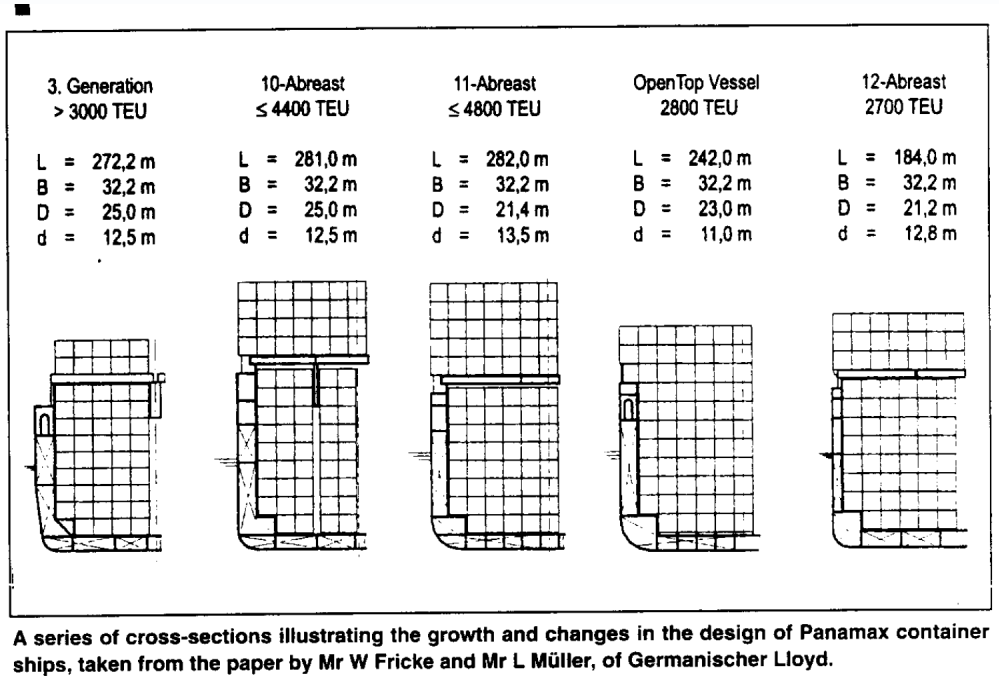
The length has stopped at about 400 m; capacity now grows with the beam, one row at a time.



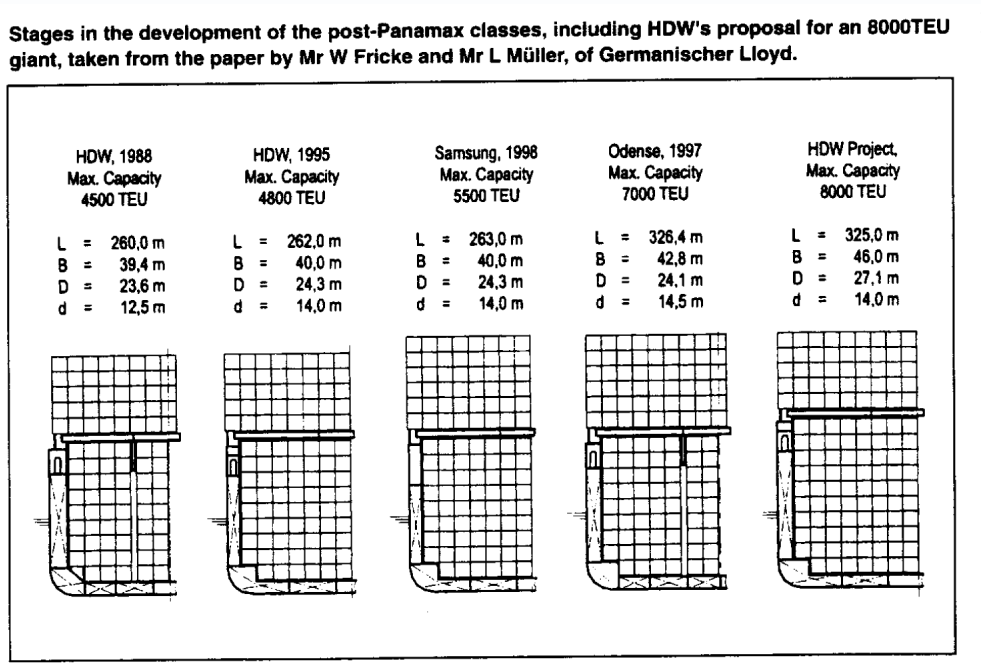
Evolution of the Main Dimensions



Evolution of the Midship Section



A series of cross-sections illustrating the growth and changes in the design of Panamax container ships, taken from the paper by Mr W Fricke and Mr L Müller, of Germanischer Lloyd.



Stages in the development of the post-Panamax classes, including HDW's proposal for an 8000TEU giant, taken from the paper by Mr W Fricke and Mr L Müller, of Germanischer Lloyd.

Panamax: 10, 11 and 12 rows across, all within B = 32.2 m

Post-Panamax: classes from 4,500 TEU (1988) to the 8,000 TEU project

Applicable Rules and the Breadth

IMO conventions

- Load Lines
- SOLAS
- MARPOL
- Tonnage
- COLREG
- Safe Containers (CSC)

Also IACS Rec. No. 45, guidelines for container corner fittings

The breadth is a multiple of the container width

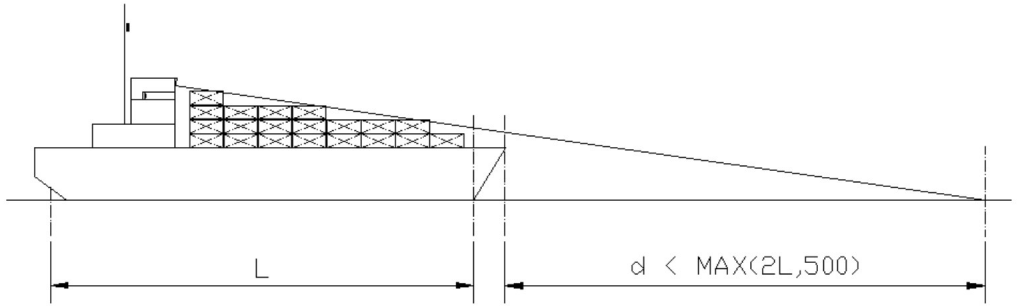
8 ft = 2.44 m per container, plus a gap d between stacks:

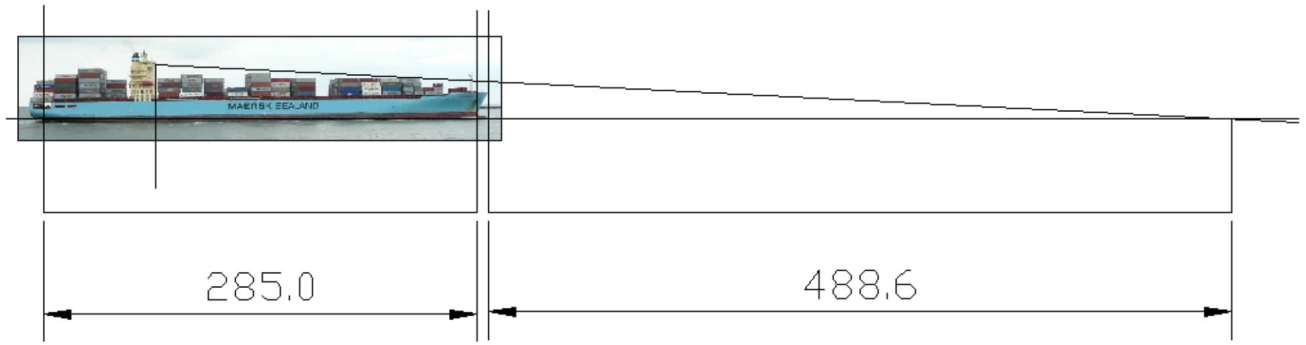
Fittings	d
standard, depending on the fittings	0.025 or 0.080 m
fixed cell guides	0.100 m

Example: B = 42.80 m for n = 17 containers across.

Visibility from the Navigation Bridge

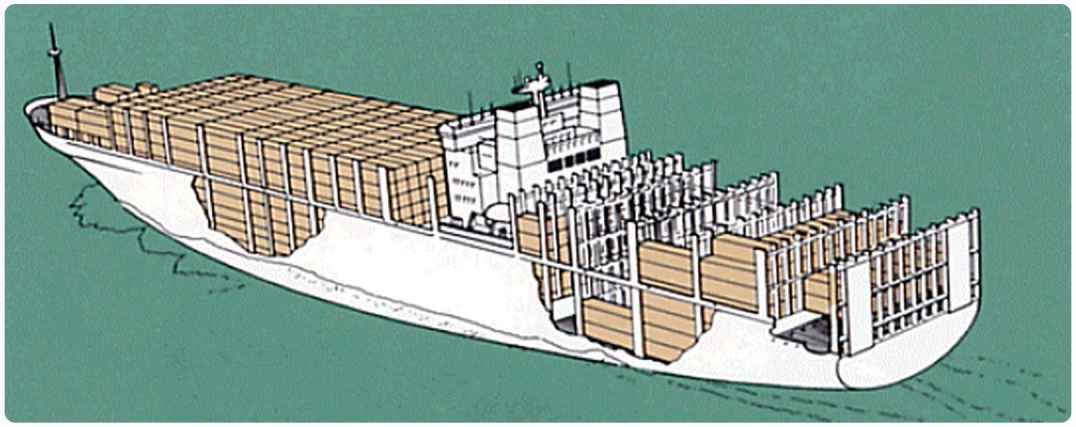
SOLAS V/22, ships over 45 m: the sea surface must be visible from no more than **2 L or 500 m, whichever is less**, ahead of the bow, within 10° to each side, in all conditions of draught, trim and deck cargo.





Example: L = 285 m, so 2 L = 570 m and the limit is 500 m; the line reaches the sea at 488.6 m: it passes.

Hatchless (Open-Top) Ships





For

- No hatch covers, no lashing: shorter port stays, lower costs
- Cell guides from the bottom to the top tier
- Lower centre of gravity, more load per stack

Against

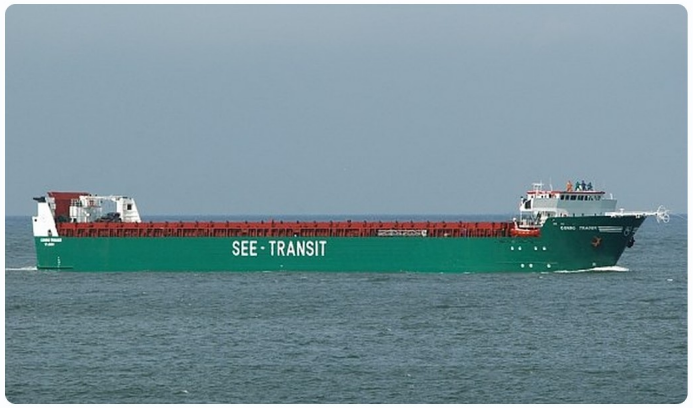
- No hazardous cargo in the open spaces above deck
- GT 50–70 % larger, price 10–12 % higher
- Containers exposed to the weather

Less Common Configurations



CONPAX

Containers and passengers. Ocean Blue Whale, 2016: 460 TEU, 800 passengers, L_OA 182.6 m, 25.2 kn.



CONRO

Containers on the upper deck, wheeled cargo below. ConRo Trader, 1978: L_OA 109.6 m, 4,550 t DW.



Arctic route

Venta Maersk, 3,600 TEU ice class: first voyage on the Northern Sea Route, August 2018.

General Design Remarks

14 t per TEU

cargo mass for the stability computations

35 % → 58 %

share carried on deck, 1970s to 2020

11 + 9 tiers

at most, in the holds and on deck

- Since 1 August 2007 the fuel oil tanks must be inside a double hull; large ships often place them across, between holds.
- Above 13,000 TEU the bridge and accommodation are likely to move forward: check the SOLAS visibility.
- Small container ships often use free-fall lifeboats, which save space at the sides.

PART B

Designing a Container Ship

Sizing from bays, rows and tiers: the method of the design manual, step by step, and the worked example.

The Story in Four Chapters

1

Why container ships are different

Containers set the hold; four typical layouts; the hull around the grid; the rules that fix the layout.

2

From TEU to hull · Steps 0 to 4

The share in the holds, targets from statistics, whole-number grids, the hull and its ratios, the deck under the sight line.

3

Weights and stability · Steps 5 to 8

Displacement and C_B, power and weights, the balance closed by the draught, GM with ballast, the five checks.

4

Worked example

A 596 TEU feeder designed again from its capacity: 20 candidates, three through the sweep, one acceptable design.

Containers Set Every Dimension of the Hold

Container	Length	Width	Height: standard / high cube	TEU
20 ft	6.058 m	2.438 m	2.591 m / 2.896 m	1
40 ft (FEU)	12.192 m	2.438 m	2.591 m / 2.896 m	2

Whole containers

A hold is a whole number of containers in each direction, plus the clearances for the cell guides. Half a row is lost space.

Small changes, big effects

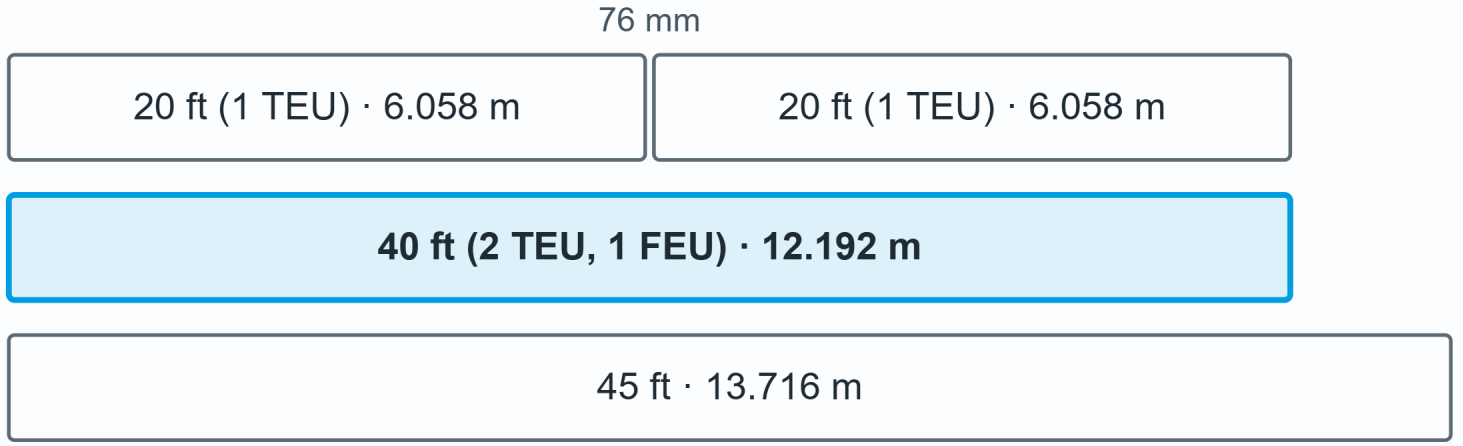
A few centimetres of beam or depth can add or lose a whole row or tier, and with it a stack in every bay.

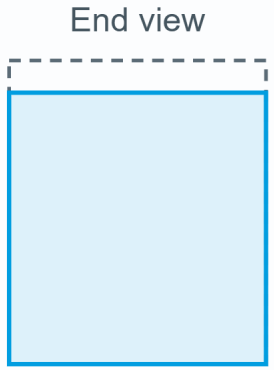
High cubes

0.305 m more per tier. Statistics from ships with 8 ft 6 in containers give too shallow a hull for high cubes.

A 40 ft container is two 20 ft containers plus 76 mm

ISO 668 containers drawn to one scale





width 2.438 m
standard height 2.591 m
high cube 2.896 m (dashed)

Maximum gross mass 36 t (ISO 668:2020)
Design mass for the project: 10 t per TEU; homogeneous 14 t per TEU

Bays, Rows and Tiers

The grid in the holds

- **n_R rows** across the beam
- **n_T tiers**, one above the other
- **n_B bays** along the length, each one 20 ft container long

A cell is one slot; a stack is one row in one bay. A hold, between two bulkheads, takes one 40 ft bay, so n_B is even; larger ships often take two 40 ft bays per hold.

In the holds, N_H: the hull limits

Beam and depth set the rows and tiers; the hull narrows at the ends.

On deck, N_D: the ship limits

Usually two rows wider than the hold, n_TD tiers on the hatch covers. Stability, lashing and the view from the bridge cap them.

First decision: split the owner's N_TEU into N_H and N_D. The share in the holds, s_H = N_H / N_TEU, is not given; it is swept.

35 % → 58 %
share on deck, 1970s to 2020

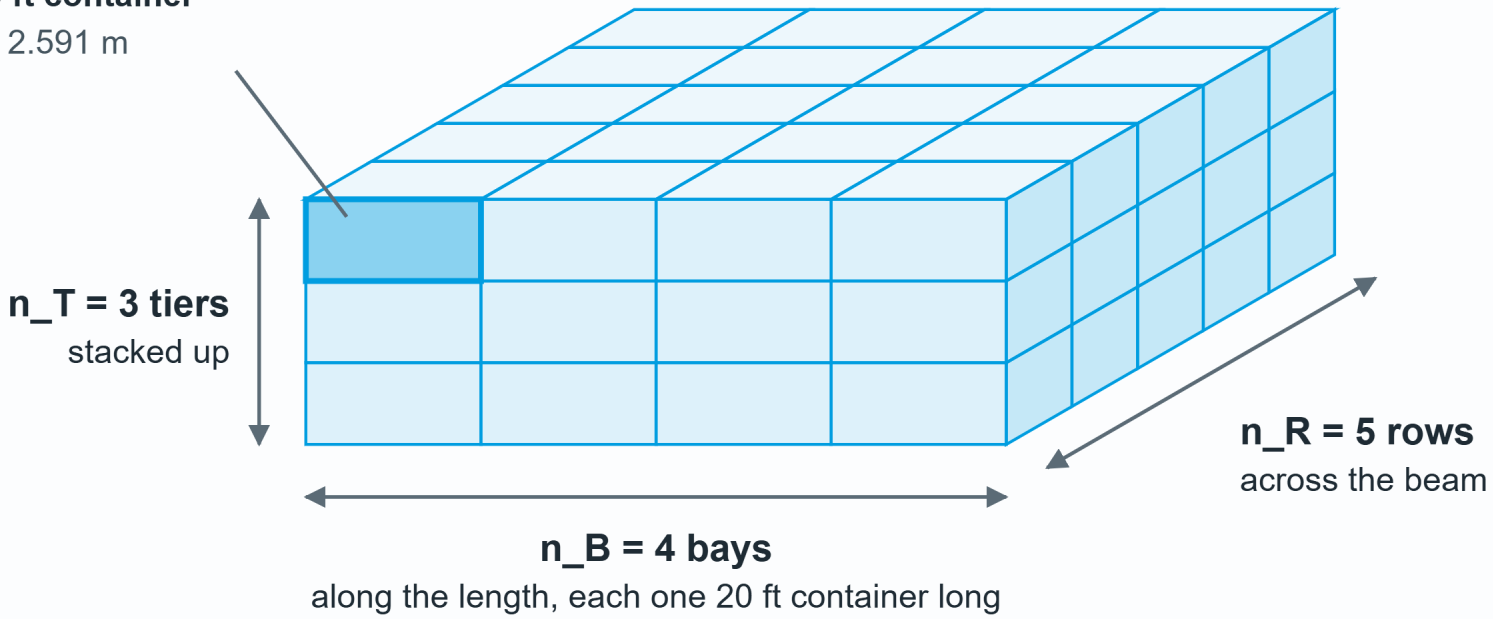
11 + 9 tiers
at most, in the holds + on deck

14 t per TEU
for stability, not the 10 t average

The grid in the holds: n_B bays × n_R rows × n_T tiers

Example of a box-shaped hold with 4 bays, 5 rows and 3 tiers; schematic

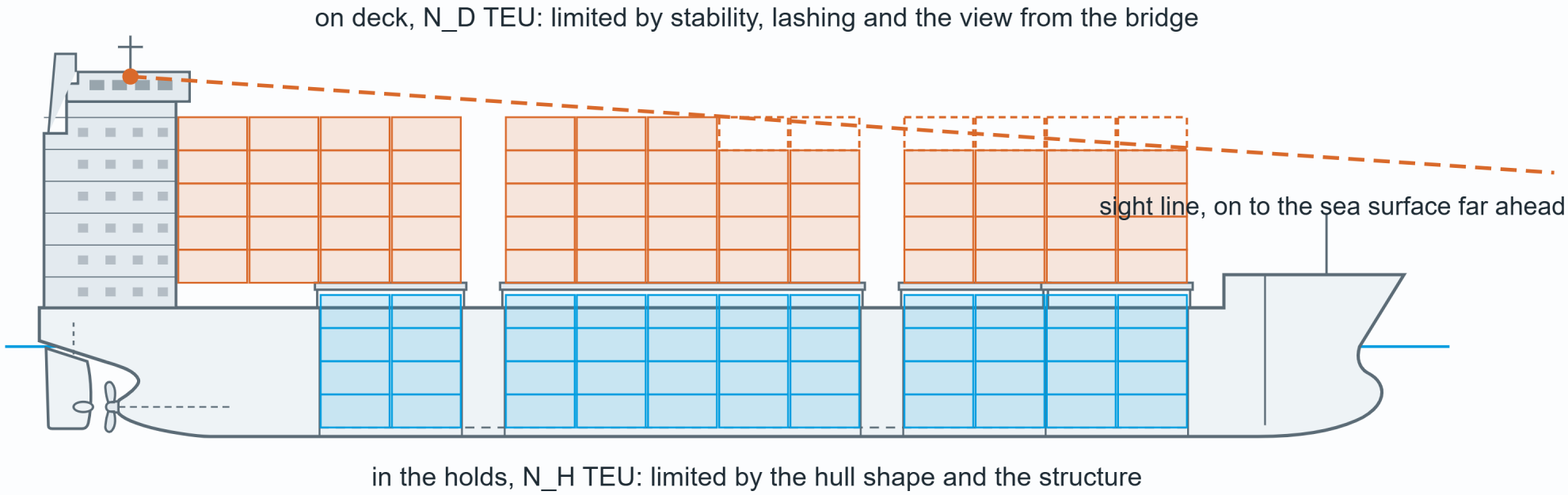
1 TEU = one 20 ft container
6.058 × 2.438 × 2.591 m



Box-shaped hold: n_B × n_R × n_T = 4 × 5 × 3 = 60 TEU. A real hull keeps only a share k_U of the cells (I.3).
Reference ship: n_B = 11, n_R = 6, n_T = 4, so 264 cells; the hull holds 228 TEU, so k_U = 0.86.

Holds and deck are limited by different things

Reference ship, to scale; a fifth deck tier is drawn to show that the sight line cuts the forward stacks first (dashed)



The Hull Around the Containers

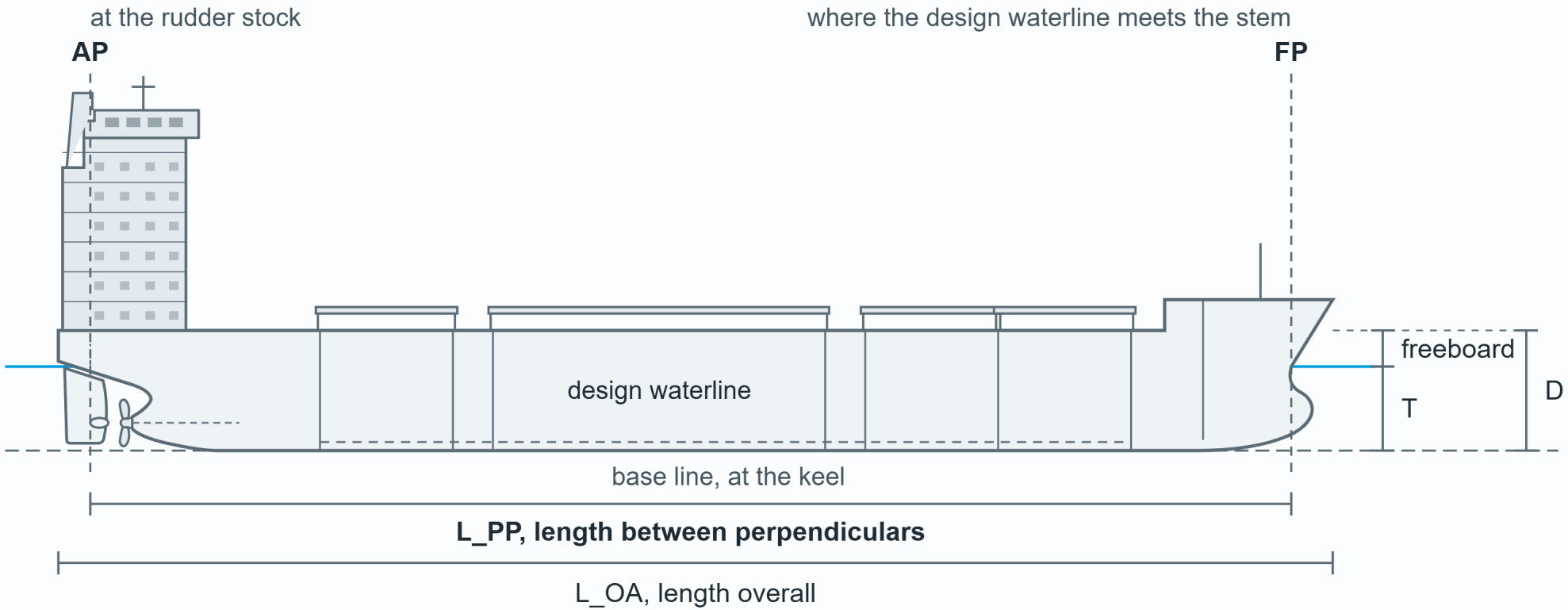
Allowance	Symbol	Typical value	Reference ship
Double bottom	h_DB	1.5 m to a little over 2 m	0.8 m
Wing tanks, each side	w_DH	2.0–2.5 m, about 0.1 B	2.1 m
Hatch coaming	h_CO	from a reference ship	1.6 m
Margin under the cover	c_HC	4–5 % of the cover; room for its deflection	0.49 m
Guide clearances across	g_BC, g_BF	0.15–0.60 m; 0.08–0.10 m	0.15, 0.08 m

$$B = 2\,w_{DH} + 2\,g_{BC} + n_R\,b_c + (n_R - 1)\,g_{BF} \quad (10)$$

$$D = n_T\,(h_c + 0.013) + h_{DB} + c_{HC} - h_{CO} - c_{DK} \quad (11)$$

Main dimensions are measured from the perpendiculars and the base line

Profile of the reference ship, to scale; the deck at the side sets D



The Hull Narrows at the Ends

The full grid n_B · n_R · n_T fits only the parallel middle body. Forward and aft, the hull loses rows and tiers. One number, the hold utilization, measures what is left:

$$k_U = N_H / (n_B \cdot n_R \cdot n_T)$$

Without a better statistic: $k_U = 0.5\, C_B + 0.4$ (8)

0.74
formula, C_B 0.68 assumed

0.86
reference ship, 228 TEU in 264 cells

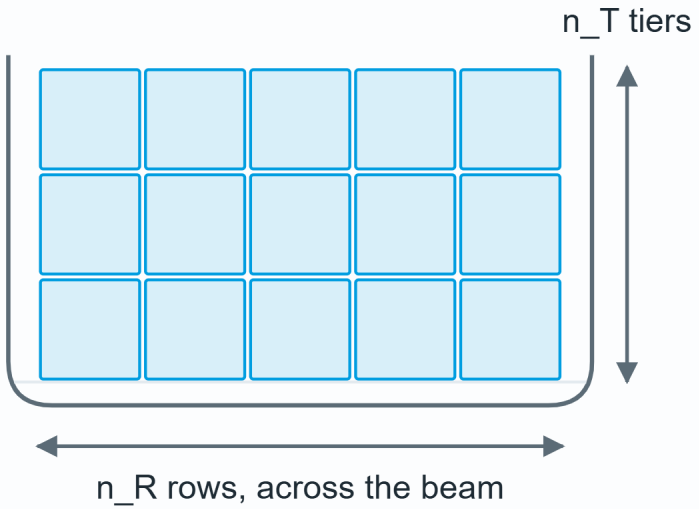
What it hides

- A bay holds only what fits at its narrower end.
- Where the inner hull just misses the minimum distance to the shell, a small change of the hull or the wing tanks gains a stack.
- k_U sets the number of bays, and so the length: calibrate it on a reference ship.
- Once hull lines exist, count the containers in the hull instead.

The hull holds only a share k_U of the full container grid

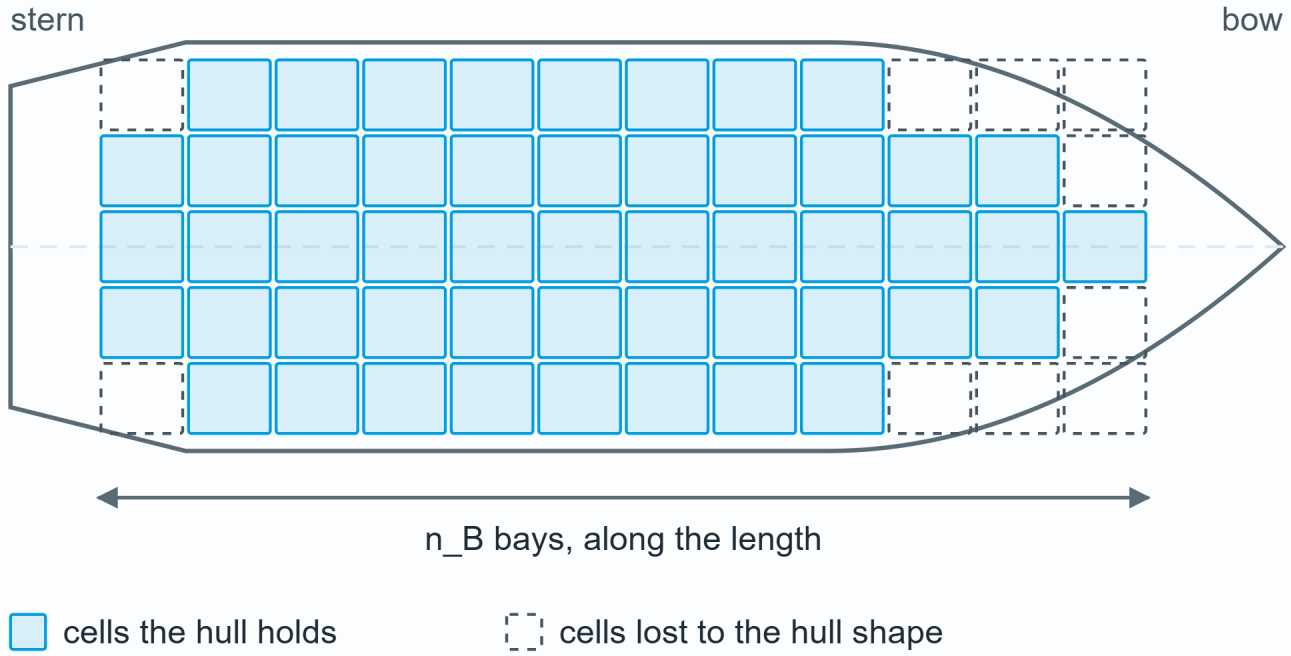
Schematic, not to scale; one cell is one 20 ft container (1 TEU)

Section: rows across, tiers up



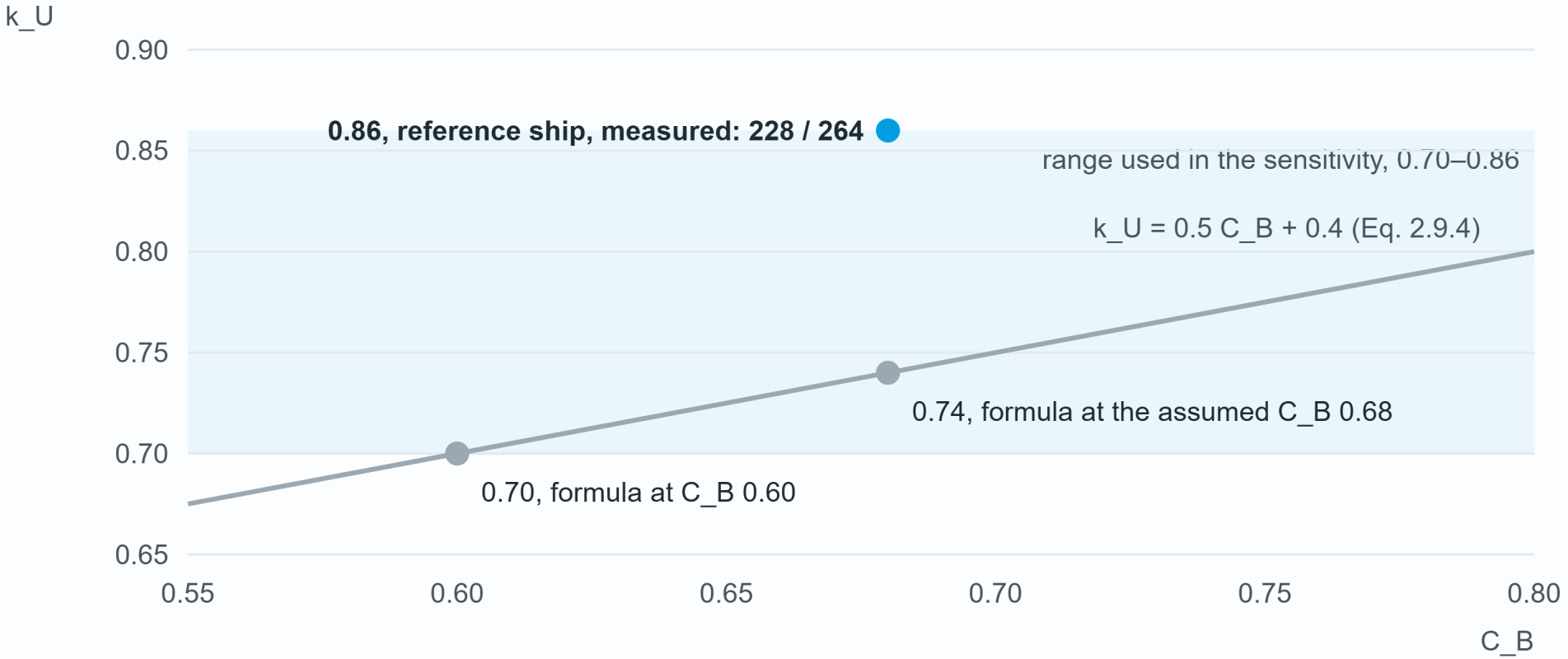
k_U = cells held / (n_B × n_R × n_T)

Plan: bays along the length



The formula gives 0.74 for the reference ship; the real ship carries 0.86

Hold utilization k_U against the block coefficient C_B



Reference ship: no estimate of the hold capacity is within 10 %

TEU in the holds, 6 rows × 4 tiers × 11 bays, C_B 0.68 assumed, against the real ship



Rules That Fix the Layout

Who	What it requires	What it fixes
Owner’s brief	N_TEU, deadweight, speed, mass per TEU	the inputs, Step 0
Class society	minimum double-bottom height and wing-tank width	the allowances around the grid
SOLAS II-1/12	collision bulkhead 0.05 L or 10 m (the less) to 0.08 L or 0.05 L + 3 m (the greater) aft of the FP	the fore peak, the forward end of the holds
SOLAS V/22	sea visible from the bridge within 2 L_OA or 500 m (the less) ahead of the bow	the deck tiers, most at the bow
Load Lines, ICLL	a minimum freeboard	D – T
Canals and ports	locks and depths: 32.2 m for the original Panama locks	B and T

The Method in Nine Steps

Step	What it decides	Main result
0 · Inputs	what the brief asks and the reference ship gives	N_TEU, V, DWT, allowances
1 · Split and targets	share in the holds, swept 0.30–0.50; targets	N_H, N_D, B_0, D_0
2 · Container grids	four grids around the targets; bays	n_R, n_T, n_B
3 · Dimensions	hull from the grid; ratios of built ships	L_PP, B, D
4 · Deck and visibility	tiers on deck under the sight line	n_TD
5 · Displacement and form	C_B and draught that float the ship	Δ, C_B, T
6 · Weights	power, lightship, deadweight in balance	P, LWT, T
7 · Stability	KG, GM with ballast, roll period	GM, T_φ
8 · Checks	keep the candidates that pass all five checks	acceptable designs

Step 1 · Split the TEU, Then Set Targets

$$N_H = s_H \cdot N_{TEU},\ N_D = N_{TEU} - N_H,\ s_H = 0.30 \ldots 0.50 \quad (1)$$

Regressions of built ships up to Panama beam, from the TEU in the holds (2), (3):

$$L_0 = 113.5 + 0.082\, N_H - 3607.5 / N_H$$

$$B_0 = \exp(0.674 - 2.7 \cdot 10^{-4}\, N_H + 0.44 \ln N_H)$$

$$D_0 = \exp(0.249 - 5.36 \cdot 10^{-6}\, N_H + 0.375 \ln N_H)$$

$$T_0 = -6.05 + 2.49 \ln N_H \quad \text{(freeboard draught)}$$

They are targets, not dimensions. Sweep s_H in steps of 0.05; a reference ship shows where in the range to look.

596 TEU at $s_H = 0.40$

Quantity	Value
N_H; N_D	238; 358 TEU
L_0	117.9 m
B_0	20.44 m
D_0	9.97 m
T_0	7.58 m

Cross-check: Papanikolaou Fig. A.47 gives 116.0 m for 596 TEU; L_0 is 1.6 % above it.

Step 2 · Whole Numbers Around Each Target

$$n_R = \lfloor (B_0 - 2\,w_{DH}) / 2.54 \rfloor,\quad n_T = \lfloor (D_0 + c_{DK} + h_{CO} - h_{DB} - c_{HC}) / (h_c + 0.013) \rfloor \tag{7}$$

$$n_B \cdot n_R \cdot n_T \cdot k_U \geq N_H,\ n_B \text{ even} \tag{8}$$

At $s_H = 0.40$ the targets give **6.39 rows** and **3.53 tiers**. Rounding down loses up to one row and one tier, and the targets are only statistics.

So keep the whole numbers on **both sides** of each target: two choices for rows times two for tiers, **four grids per share**. The checks decide.

Grid	Rows × tiers	Bays	$k_U = 0.74$
1, as rounded	6 × 3	18	238 / (18 · 0.74)
2, one more tier	6 × 4	14	238 / (24 · 0.74)
3, one more row	7 × 3	16	238 / (21 · 0.74)
4, both	7 × 4	12	238 / (28 · 0.74)

Step 3 · Build the Hull, Then Check Its Ratios

> $L_{PP} = L_{AP} + L_{ER} + L_{FP} + L_C$, with $L_C \approx 6.55\, n_B$ (9); B and D from eqs. (10) and (11)

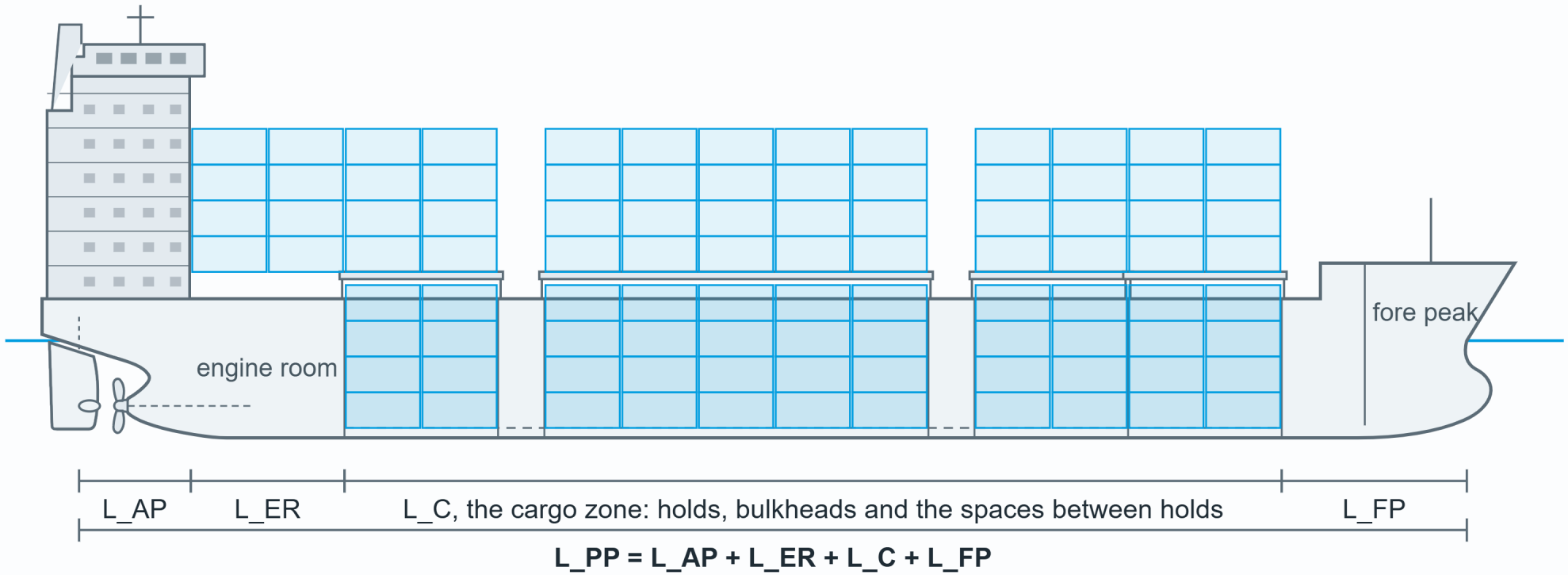
Ranges of built Panama ships with fewer than 1,000 TEU in the holds: L/B 5.2–6.5 · B/D 1.7–2.2 · B/T 2.4–3.2 · L/D 10.5–13.0 · T/D 0.63–0.79 · Fn 0.23–0.26

$s_H = 0.40$	$L_{PP} \times B \times D$	Ratios outside the range
6 × 3 × 18	154.6 × 19.53 × 8.42 m	L/B 7.92, B/D 2.32, L/D 18.4, T/D 0.90
6 × 4 × 14	128.4 × 19.53 × 11.33 m	L/B 6.58: too slender
7 × 3 × 16	141.5 × 22.05 × 8.42 m	B/D 2.62, L/D 16.8, T/D 0.90
7 × 4 × 12	115.3 × 22.05 × 11.33 m	L/D 10.18: too short for its depth

No grid of this share passes. A failed grid is dropped, not repaired: its fix is another grid or share that the sweep already contains.

Length is the cargo zone plus the spaces that carry no containers

Profile of the reference ship, to scale



Step 4 · Deck Tiers Under the Sight Line

> $N_D \le (n_B + n_l)(n_R + 2)\, n_{TD}$, $n_{TD} \le 5$ (12)
>
> Sight line from the eye to the sea at 2 L_{OA} or 500 m ahead of the bow; each bay keeps the tiers under it (13)–(15)

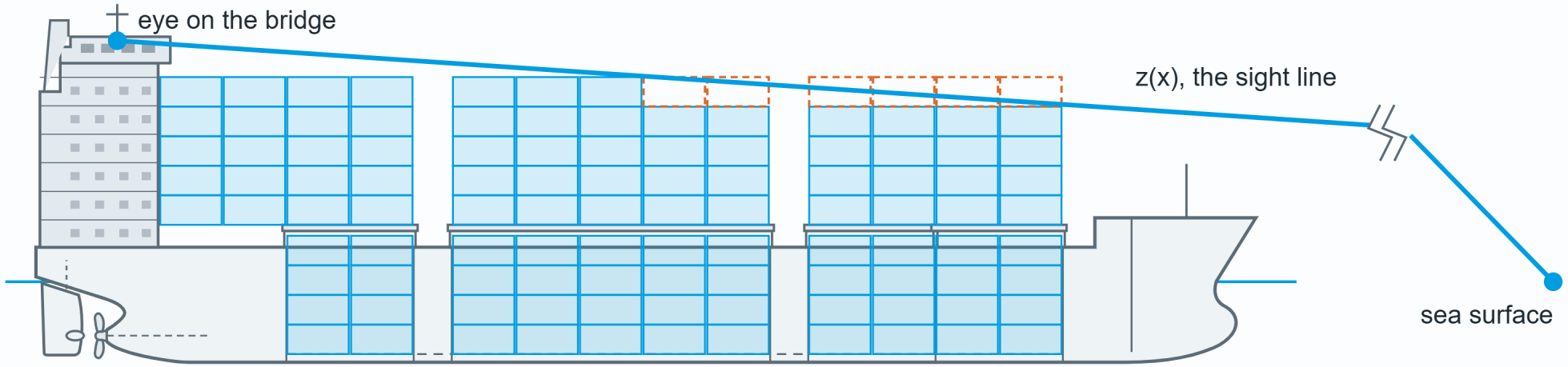
Eq. (12) puts the same tiers on every bay; the sight line does not. Apply it bay by bay, the foremost bay first.

Example, 7 × 4 × 14 at $s_H = 0.45$: $N_D = 328$ needs $n_{TD} = 3$, which carries 432 TEU. Eye 27.9 m above base, 10.8 m from the aft perpendicular: every bay keeps 3 tiers.

Next slide: the sight line on the reference ship, from the design manual (Figure II.9).

Deck stacks must stay below the line from the eye to the sea

Reference ship, to scale up to the break, with a fifth tier drawn; beyond the break the line runs on to the sea surface



filled: tiers below the line; dashed: tiers above it, not allowed

Step 5 · C_B and Draught, Solved Together

$$C_B = 0.8217\, L_{PP}^{0.42}\, B^{-0.3072}\, T^{0.1721}\, V^{-0.6135}, \quad \Delta = 1.030\, C_B\, L_{PP}\, B\, T \tag{19}$$

$$C_M = 1 - 0.062\, Fn^{0.792}, \quad C_P = C_B / C_M \tag{20}$$

L_PP and B are fixed by the grid. Katsoulis' C_B needs T, and T needs C_B, so loop:

1. start from any draught, for example B / 2.9;
2. C_B from Katsoulis at that draught;
3. the draught that gives Δ with that C_B;
4. repeat until T changes by less than 1 cm.

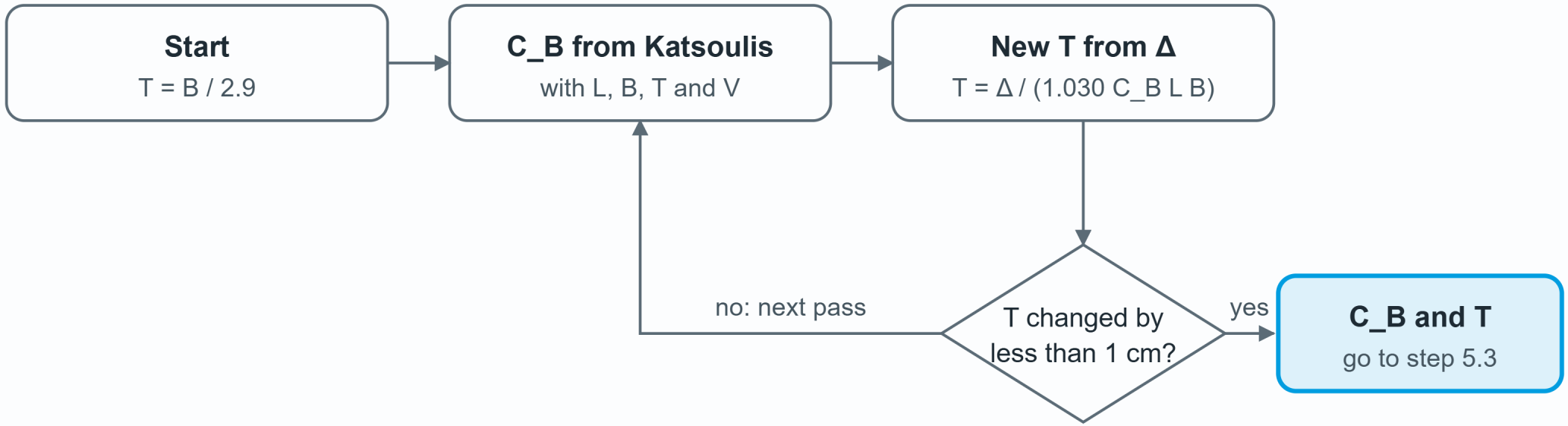
The exponent of V is negative: a slower ship is fuller. Use the service speed of the brief.

Reference ship hull, Δ = 12,494 t (C_B 0.68 at 7.9 m, assumed), at 17 kn:

Pass	T in	C_B, T out
1	6.897 m (B / 2.9)	0.584, 9.194 m
2	9.194 m	0.614, 8.750 m
3	8.750 m	0.609, 8.825 m
4	8.825 m	0.610, 8.812 m
5	8.812 m	0.609, 8.815 m

C_B and T are solved together, usually in three to five passes

L and B fixed by the grid, Δ from the weights, V from the brief



Step 6 · Power and Weights, Group by Group

Group	Method (Papanikolaou §2.15)	First candidate
Power P	Admiralty: P = Δ^(2/3) V³ / C_N, with C_N = 403 from the reference ship	6,148 kW
Consumables	fuel at 190 g/kWh, auxiliaries 6 %, reserve 1.3; lub oil 4 %; water, provisions, crew. Range 3,000 nm and 16 crew assumed	324 t
Hull steel	Schneekluth for container ships, C′_ST = 0.093; includes the cell guides	2,306 t
Deckhouse	Müller–Köster, five layers, full beam, 12 m long	203 t
Outfit	0.36 t/m² of L · B, hatch covers included	1,019 t
Machinery	Watson and Gilfillan: 12 P / n + 0.69 P^0.70, 500 rpm	457 t
Lightship	the above plus a 2 % margin R; KG from each group’s height	4,065 t, KG 7.81 m

Admiralty is a first estimate for comparing candidates. Power the chosen design with a resistance and propulsion method, Holtrop and Mennen, then close the balance again.

Close the Balance by the Draught

$$\Delta = 1.030\, C_B\, L\, B\, T = \mathrm{LWT} + \mathrm{DWT} \quad (30)$$

Power and lightship from the draught, eq. (19) for a new C_B and T; repeat until T moves less than 1 cm.

Pass	T, C_B in	P	LWT + DWT	Hull, 1.030 C_B L B T	Shortfall	T, C_B out
1	7.60 m, 0.680	6,799 kW	13,097 t	15,072 t	−1,976 t	7.30 m, 0.616
2	7.30 m, 0.616	6,191 kW	12,963 t	13,097 t	−134 t	7.23 m, 0.615
3	7.23 m, 0.615	6,149 kW	12,958 t	12,963 t	−4 t	7.23 m, 0.615

First candidate, 7 × 4 × 14, DWT = 8,893 t at 16.5 kn. The hull floats more than the ship weighs, so the draught comes down. After three passes the sides differ by 4 t, against a margin R of 80 t; k_U = 0.71 still gives 14 bays.

Why Only the Draught Moves

	Deadweight carrier (Table 1.8)	Container ship (Table 1.9)
Starts from	the deadweight, then Δ	the TEU, then the grid
L, B, D	from Δ and the ratios; any value	from bays, rows and tiers; whole containers
Δ	a first estimate	a result: LWT + DWT
Draught	from B/T and L/T, early	last, from Δ, L, B and C_B
Weights change	scale the hull	change T and C_B
Ratios	used to size the hull	used to check it

A hull 3 % wider would need 7.2 rows. If the draught that floats the weights does not suit the hull, take another grid or share: the hull is not stretched.

Step 7 · Stability with the Deck Stacks

$KG = \Sigma\, m_j\, z_j / \Sigma\, m_j$, each tier at half its height (31), (32)

$GM = KB + BM - KG - \delta GM_{FS}$, $BM = I_T / \nabla$ (33)

$C_{WP} = (1 + 2\, C_B) / 3$, one slack fuel tank assumed (34)

$T_\varphi = 2\, C\, B / \sqrt{GM}$, IS Code (35)

BM grows with B^2, KG with every tier: **one more row helps stability, one more tier works against it.**

Criterion

GM ≥ 0.15 m after free surface, IS Code criterion (e), the only one without hull lines

Target, reported

GM ≥ 0.06 B, Papanikolaou §2.12

Roll period

$T_\varphi \geq 12$ s: a large GM gives violent motions on the deck stacks

Three Loading Conditions, and Ballast

Condition	Cargo	Consumables	What it gives
Design, departure	every slot at 10 t per TEU	100 %	GM, with the ballast it needs
Design, arrival	every slot at 10 t per TEU	10 %	GM, usually the critical case
Homogeneous	14 t per TEU, as many as stability and the deadweight allow	100 %	the number of TEU of 14 t

Stow the holds first

The same number of TEU in each hold tier, then the deck from the lowest tier up.

Ballast in the double bottom

In steps of 10 t until GM = 0.15 m, limited by the volume of the double bottom and the deadweight left.

Step 8 · Five Checks, Then Keep or Drop

Check	What it tests	If it fails
Freeboard	D − T at least the ICLL basic freeboard, Type B	the depth
Ratios	L/B, L/D, B/D, B/T, T/D and Fn at the balanced draught	length, beam, depth or draught
Capacity	TEU in the holds and on deck, under the sight line	another grid or share
GM	at least 0.15 m, departure and arrival, with the ballast that fits	the beam or the depth
Roll period	at least 12 s	the beam or the depth

The balance Δ = LWT + DWT is not a check: it is closed by the draught. A candidate is dropped when that draught fails the freeboard or the ratios. The candidates that pass all five are the set of acceptable designs.

A 596 TEU Feeder, Designed Again

The reference ship, as built

Particular	Value
Type	geared feeder, 2007; engine room and bridge aft
Capacity	596 TEU: 228 holds, 368 deck
L_PP × B × D	112.9 × 20.0 × 11.3 m
T; DWT	7.9 m; 8,893 t
Power; speed	6,000 kW at 500 rpm; 16.5 kn
Holds	11 bays × 6 rows × 4 tiers, high cube

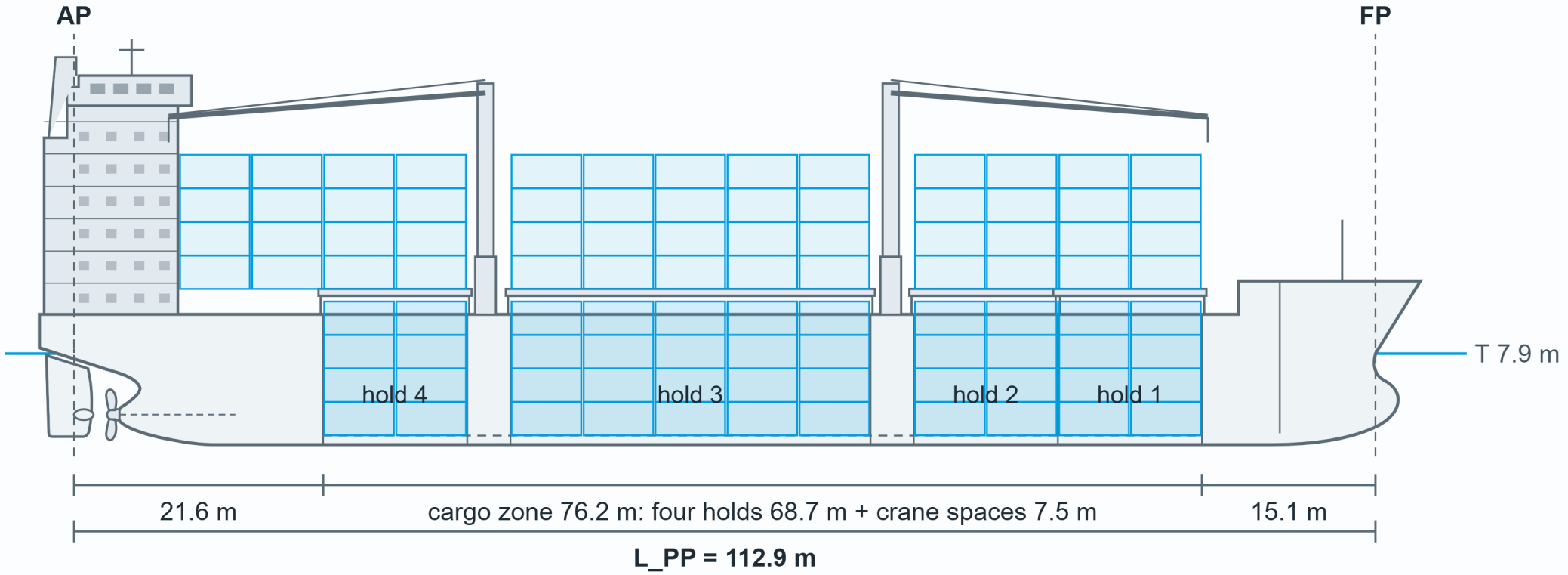
What the method takes from it

Input	Value
h_DB, w_DH, h_CO, t_cover	0.8, 2.1, 1.6, 0.6 m
L_AP + L_ER; L_FP	21.6 m; 15.1 m
Deck bays over the engine room	2
C_B for k_U	0.68, assumed
Eye above the bridge deck	1.8 m, assumed
Range; crew	3,000 nm; 16, assumed

Brief: N_TEU = 596, DWT = 8,893 t, 16.5 kn. Every number that follows comes from the worked example of the design manual (III.1).

The reference ship: 11 bays in four holds, 13 bays on deck, two cranes

Profile to scale, from the general arrangement; stacks drawn full



The Sweep: 20 Candidates, Three Pass Steps 3 and 4

s_H	1 · as rounded	2 · one more tier	3 · one more row	4 · both
0.30	5×3×18: L/B, L/D, T/D	5×4×14: L/B, B/D, T/D	6×3×14: four ratios	6×4×12: L/D, T/D
0.35	6×3×16: four ratios	6×4×12: L/D	7×3×14: B/D, L/D, T/D	7×4×12: L/D
0.40	6×3×18: four ratios	6×4×14: L/B	7×3×16: B/D, L/D, T/D	7×4×12: L/D
0.45	6×3×22: four ratios	6×4×16: L/B	7×3×18: four ratios	7×4×14: passes
0.50	7×3×20: four ratios	7×4×16: passes	8×3×18: B/D, L/D, T/D	8×4×14: passes

Rows × tiers × bays, then the ratios outside their range; “four ratios” means L/B, B/D, L/D and T/D. None fails the deck check.

> **Three tiers are too shallow** for high cubes, so those ships come out long and shallow. The real ship, 6 × 4 with 11 bays and s_H = 0.38, sits between two failing grids.

Steps 5 to 8 on the Three Candidates

	0.45, 7 × 4 × 14	0.50, 7 × 4 × 16	0.50, 8 × 4 × 14
L_PP × B × D, m	128.4 × 22.05 × 11.33	141.5 × 22.05 × 11.33	128.4 × 24.56 × 11.33
T; C_B	7.23 m; 0.615	6.68 m; 0.631	6.95 m; 0.590
Δ; P	12,958 t; 6,148 kW	13,553 t; 6,335 kW	13,339 t; 6,268 kW
LWT; its KG	4,065 t; 7.81 m	4,660 t; 7.78 m	4,446 t; 7.87 m
Ratios and Fn at T	pass	B/T 3.30, T/D 0.59, Fn 0.228	B/T 3.53, T/D 0.61
GM, no ballast, dep. / arr.	−0.24 / −0.39 m	0.58 / 0.46 m	2.04 / 1.96 m
Ballast for GM 0.15 m	740 / 1,020 t	none	none
Roll period, dep. / arr.	44.7 / 45.2 s	23.4 / 26.4 s	14.2 / 14.6 s
TEU of 14 t	528	564	596
Five checks	all pass	ratios fail	ratios fail

A higher k_U shortens the ship, and the shorter ships fail the ratio check

The three feasible candidates of the sweep: bays and L_PP (m) for four values of k_U

s_H, rows × tiers	k_U = 0.70 C_B 0.60	k_U = 0.74 formula, C_B 0.68	k_U = 0.80	k_U = 0.86 reference ship
0.45, 7 × 4	14 bays · 128.4	14 bays · 128.4	12 bays · 115.3	12 bays · 115.3
0.50, 7 × 4	16 bays · 141.5	16 bays · 141.5	14 bays · 128.4	14 bays · 128.4
0.50, 8 × 4	14 bays · 128.4	14 bays · 128.4	14 bays · 128.4	12 bays · 115.3

Darker: longer ship. Dashed: 115.3 m, where L/D = 10.2, below the 10.5–13.0 range (the 8-row ship also has L/B = 4.7).
Each step is two bays, 13.1 m of length.

One Acceptable Design, Limited by Stability

7 × 4 × 14
rows × tiers × bays, s_H = 0.45

7.23 m
draught, C_B 0.615, on 128.4 × 22.05 × 11.33 m

740 t
ballast for GM 0.15 m on departure; 1,020 t on arrival

528 TEU
of 14 t, limited by stability

Against the real ship	Design	Reference ship
L_PP × B	128.4 × 22.05 m	112.9 × 20.0 m
Draught; power	7.23 m; 6,148 kW	7.9 m; 6,000 kW
TEU of 14 t	528	410, published

Before it is accepted: power it by Holtrop and Mennen and close the balance again; correct the freeboard; replace the assumed range, crew and deckhouse by the owner’s.

The Story in Four Lines

1 · Containers set the hull. Whole rows, tiers and bays plus the allowances give L, B and D; the hull narrows at the ends, k_U measures it.

2 · Sweep, do not choose. Five shares in the holds times four grids around the targets: 20 candidates, checked against built ships.

3 · The draught closes the balance. With the grid fixed, only T and C_B move; a draught that does not suit the hull means another grid.

4 · Stability decides. For the 596 TEU brief one design passes; it needs ballast, and stability, not space, limits its TEU of 14 t.

For your project: the design manual has every step and the worked example behind every number here. Repeat it with your own brief and reference ship.